Big Vision
比格威医疗科技

BV1000S

New generation fully automated artificial intelligence OCT

A new generation of fully automatic artificial intelligence OCT

www.bigvisiontech.com



Artificial intelligence medical device standardization technology centralized member unit

BV1000S

Optical coherence tomography scanner

The BV1000S Optical Coherence Tomography for Ophthalmology is an integrated automatic artificial intelligence OCT system with both soft and hard capabilities, boasting complete independent intellectual property rights. It combines cutting-edge multi-image registration enhancement technology with industry-leading analysis techniques to offer a comprehensive solution for clinical diagnosis and treatment. This includes automated image capture, precise image analysis, intelligent and real-time cloud-based data sharing.

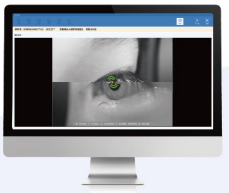




Proficient in fundus imaging

Assist in improving the reliability and effectiveness of clinical diagnosis

Product Features

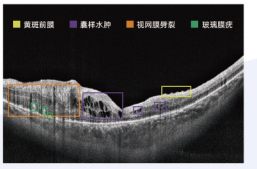


Fully automatic operation

High-resolution imaging

Fast pupil positioning and focusing; Full voice guidance; It takes an average of 2 minutes to complete the examination of both eyes of one patient.

12mm×9mm Large field scan; It covers both the macular area and the optic disc area. Retinal disorders can be comprehensively detected



AI Image Analysis

Industry-leadingimage algorithms

It can be used for auxiliary diagnosis of 16 main fundus lesions. The lesion area was accurately identified, segmented and automatically marked.

Provide reliable and accurate quantitative analysis(such as mean choroidal thickness, central macular thickness, etc.)



Intelligent eye health management

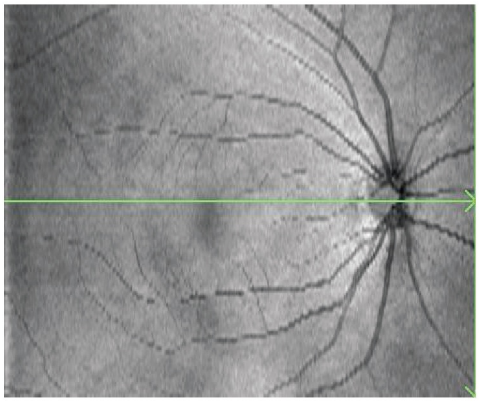
Perfect adaptation to all kinds of scenarios

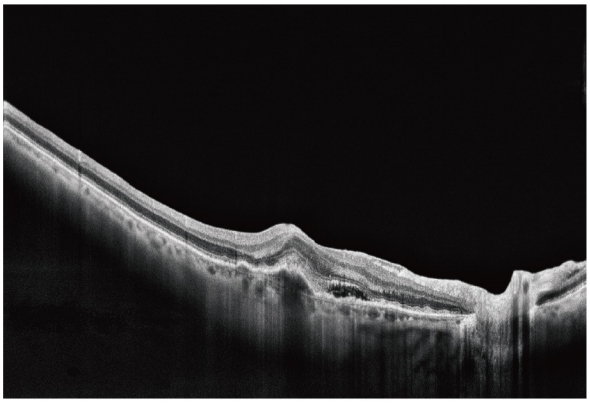
Automatically establish eye health records for users; Mobile scanning code can quickly receive the inspection report.

Using cloud architecture; Real-time sharing of examination image data; To provide platform support for remote hierarchical medical alliance

Ultra-fast fundus signal acquisition

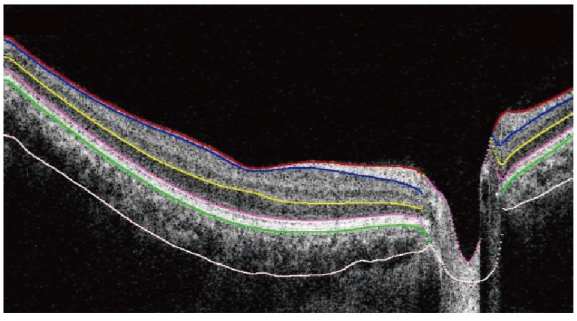
80,000 times per second "second" acceleration technology, the scanning speed is greatly improved, helping to accurately capture different subdivision forms of fundus lesions in a shorter time and obtain high-qualityimages.





Hyperfine fundus structure imaging

The AI multimodal ophthalmic image analysis algorithm based on deep learning can more finelyobserve the subdivision morphology of various fundus diseases and the development of lesions indifferent structural layers of the retina, realize the hierarchical imaging of the retina and choroid, support the monitoring, diagnosis and hierarchical analysis of the disease, and help clinicians find more isease evidence for patients and guide treatment.



Layering imaging of the retina and choroid in automatic layering mode

Fundus retinal choroidal thickness can be quantified

Ultra-portable and humanized design

The ergonomic fully automatic soft and hard artificial intelligence OCT brings users a new fully automatic equipment experience.

It adopts the split design of optical host and display, combining excellent optical design, automatic algorithm, parallel high-speed computing framework and high ease of use human-computer interaction interface. The imaging quality, analysis accuracy, calculation speed and operation convenience are greatly improved.

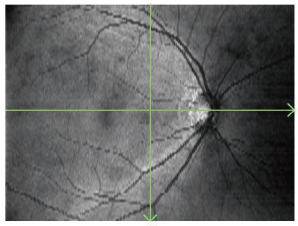


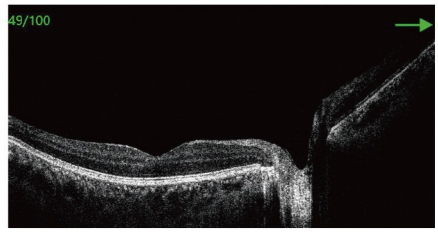
Large Field Analysis Function

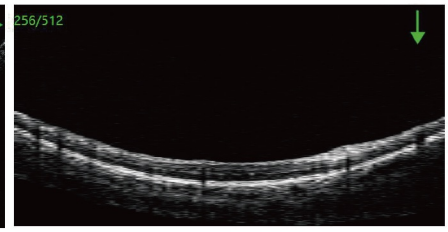
Large field scan （12mm X 9mm）

Covering the macular area and the optic disc area at the same time, can fully detect the retinal disease.

OD Signal 90



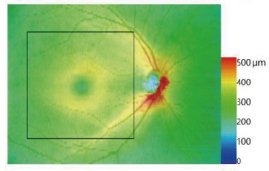




Average choroidal thickness(μm)(73~319)	254
Average retinal thickness(μm)(154~362)	286
Macular foveal thickness(μm)(142~300)	222
Total retinal volume(mm³)(4.00~12.00)	9.03

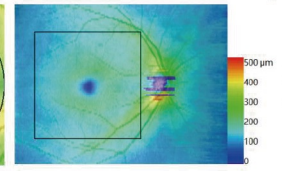
N/A indicates that it could not be analyzed

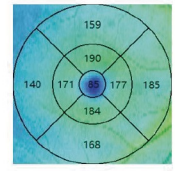
Retinal Thickness Map ILM-RPE/ETDRS



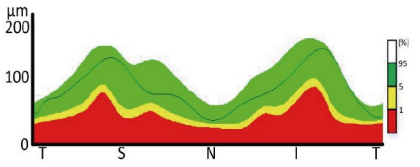


Inner retinal thickness map ILM-OPL/ETDRS (μm)

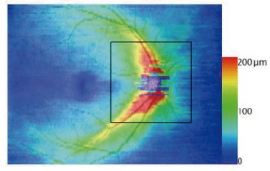


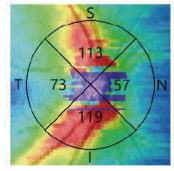


RNFL thickness diameter (3.4mm)

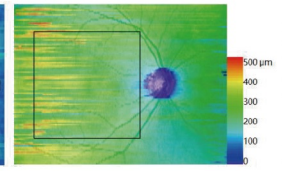


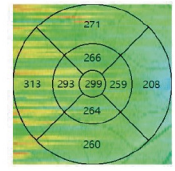
RNFL thickness map /TSNI (μm)





Choroidal thickness map BRM-SCL/ETDRS (μm)

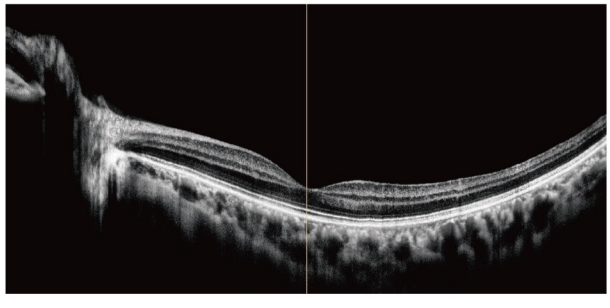




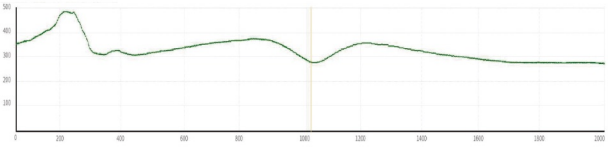
Software analysis

- 12*9mm fundus map
- Average choroidal thickness
- Macular foveal thickness
- Grid map of TSNI thickness of the nerve fiber layer
- Total retinal volume
- RNFL circular thickness
- Average retinal thickness
- Grid map of macular 6*6mmILM-RPE thickness
- Topographic map of RNFL thickness
- ILM-RPE thickness topogram
- Grid map of macular 6*6mmILM-OPL thickness
- BRM-SCL thickness topographic map
- ILM-OPL thickness topographic map
- Meshes of macular 6*6mmBRM-SCL thickness

High definition single line scan

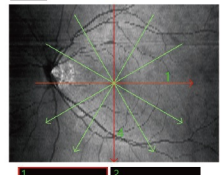


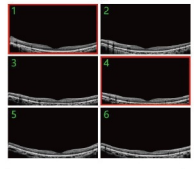
Retinal thickness (ILM-RPE layer) (μm)



Six radial lines scan

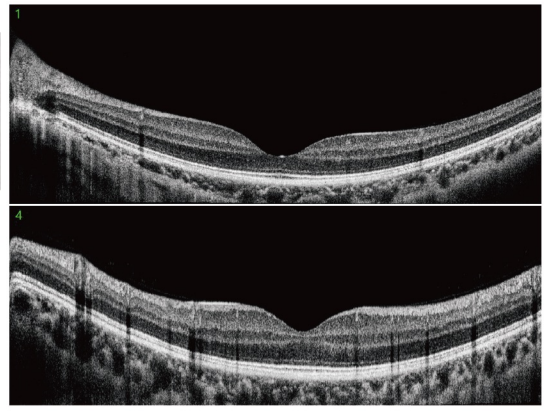
OS Signal 94





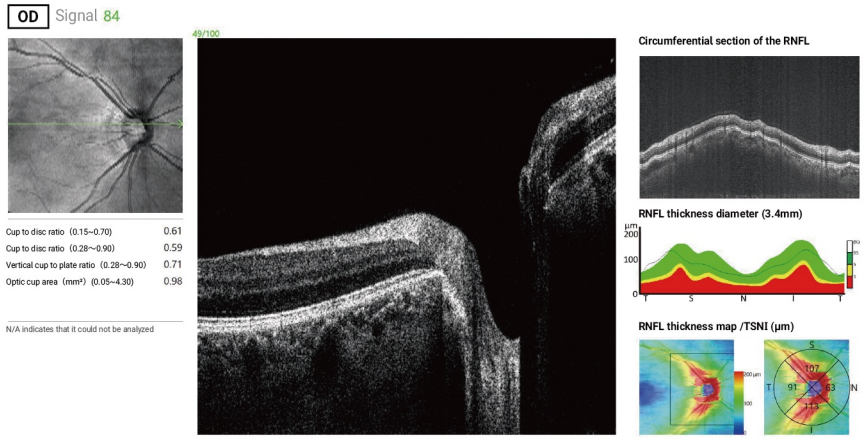
Average retinal thickness(μm)(154~362)	304
Macular foveal thickness(μm)(142~300)	196

N/A indicates that it could not be analyzed



Glaucoma analysis function

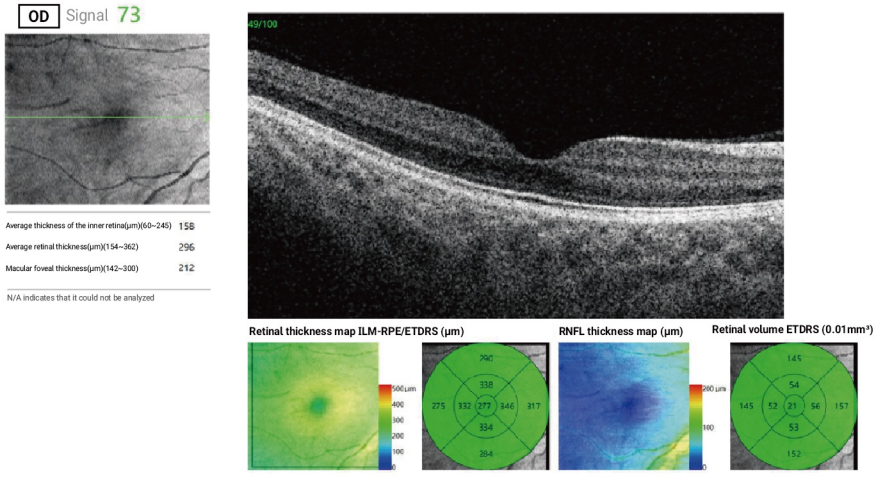
Disc area



Software analysis

- 6*6mm fundus map of the optic disc
- Cup to disc ratio
- Cup to disc ratio
- Vertical cup to plate ratio
- Optic cup area
- RNFL rim sweep thickness curve
- Topographic map of RNFL thickness
- Grid map of TSNI thickness of the nerv fiber layer

The macular area

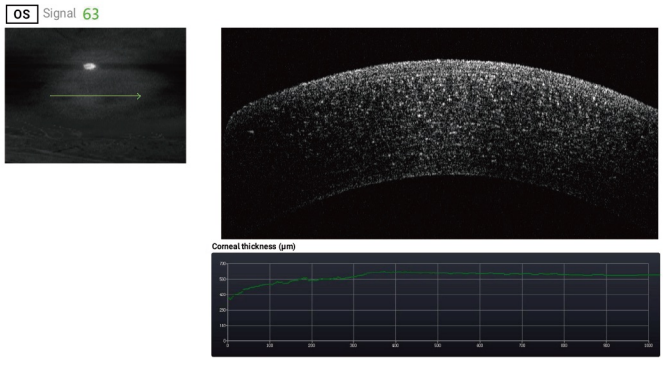


Software analysis

- 6*6mm fundus map of the macula
- Average thickness of the inner retina
- Average retinal thickness
- Macular foveal thickness
- ILM-RPE thickness topography
- Grid map of ILM-RPE thickness
- Topographic map of RNFL thickness
- Topographic map of RNFL thickness

Anterior segment analysis function

High definition single line scan of anterior segment



Six radial lines scan of anterior segment

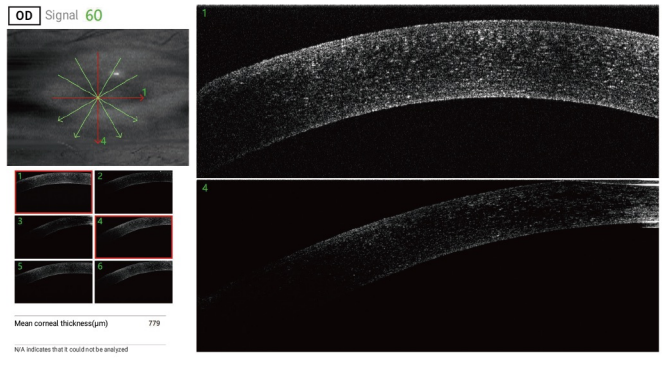


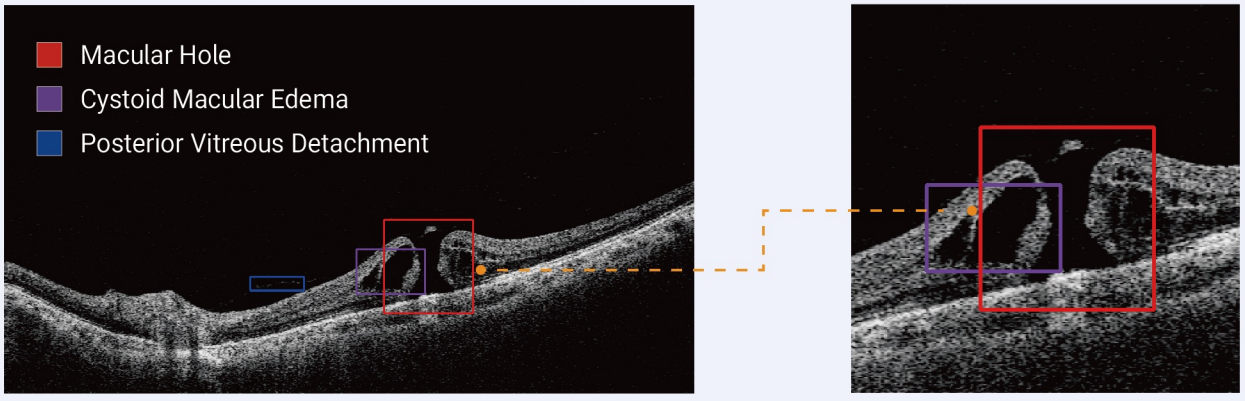
Diagram of the lesion

It can be used for auxiliary diagnosis of 16 main fundus diseases

label	Disease Name		label	Disease Name		label	Disease Name	
1	Retinal Pigment Epithelium Detachment		7	Retinoschisis		13	Elliptical Band Missing	
2	Posterior Vitreous Detachment		8	Cystoid Macular Edema		14	Choroidal Excavation	
3	Epiretinal Membrane		9	Exudation		15	Choroidal Atrophy	
4	Subretinal Fluid		10	Macular Hole		16	Retinal Hemorrhage	
5	Choroidal Neovascularization		11	Retinal Detachment				
6	Drusen		12	Ametropia				

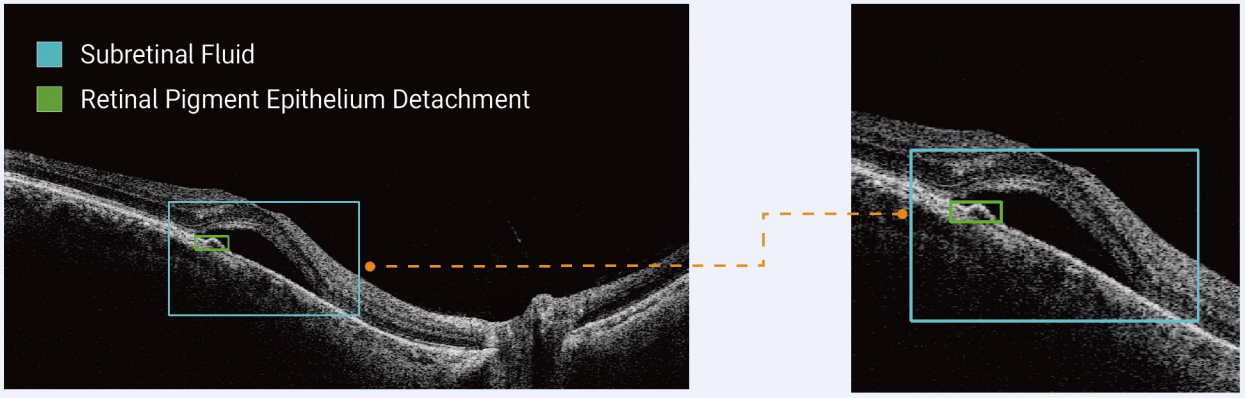
Macular Hole

Macular hole is a tissue loss of the entire layer or part of the macular nerve retina. Most of them are idiopathic and related to abnormal vitreomacular traction, and a few are related to trauma. Lamellar macular hole (LMHH) is a layer of tissue loss on the surface of the fovea and is often associated with the epiretinal membrane.

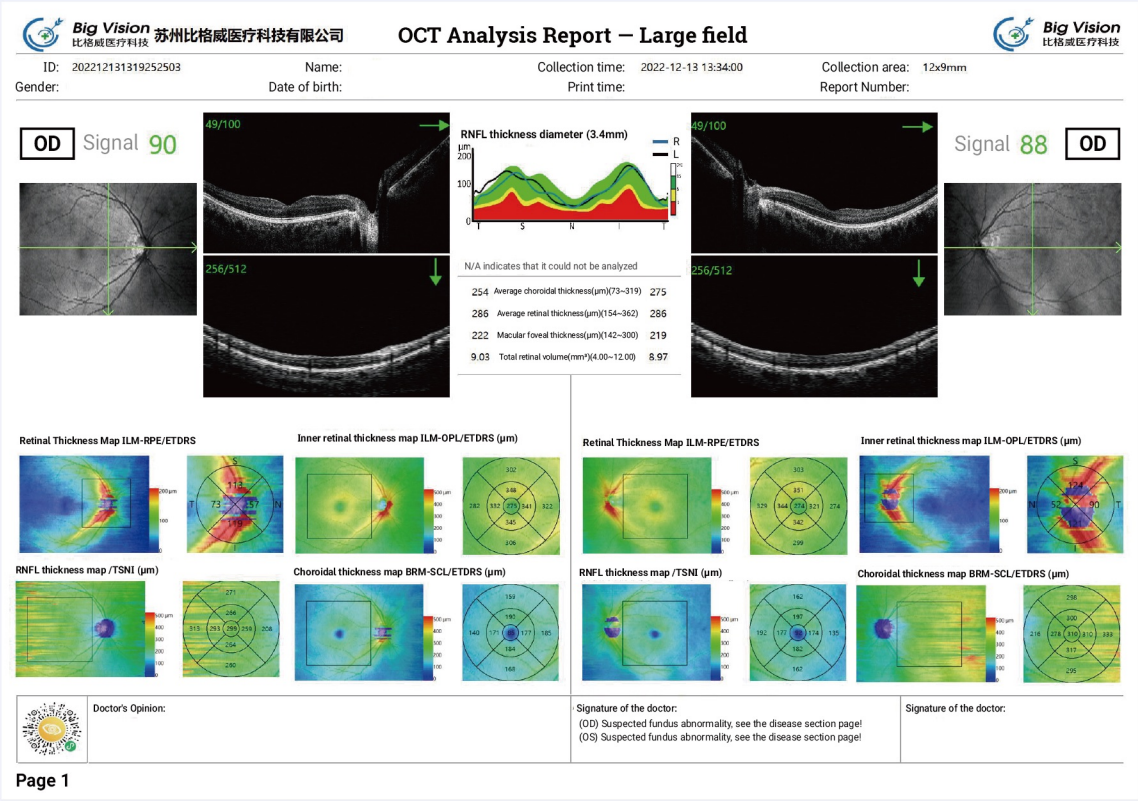


Subretinal Fluid

On OCT, there was a serous detachment under the retinal neuroepithelial layer, with uniform hyporeflection within the detachment area, and a continuous light reflection zone in the retinal pigment epithelial layer below.



Report of examination



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OCT Analysis Report — Large field

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ID: 202212131319252503 Name: Collection time: 2022-12-13 13:34:00 Collection area: 12x9mm

Gender: Date of birth: Print time: Report Number:

OD Signal 90

Signal 88 OD

RNFL thickness diameter (3.4mm)

N/A indicates that it could not be analyzed

254	Average choroidal thickness(μm)(73~319)	275
286	Average retinal thickness(μm)(154~362)	286
222	Macular foveal thickness(μm)(142~300)	219
9.03	Total retinal volume(mm³)(4.00~12.00)	8.97

Retinal Thickness Map ILM-RPE/ETDRS

Inner retinal thickness map ILM-OPL/ETDRS (μm)

RNFL thickness map /TSNI (μm)

Choroidal thickness map BRM-SCL/ETDRS (μm)

Doctor's Opinion:

Signature of the doctor:

(OD) Suspected fundus abnormality, see the disease section page!

(OS) Suspected fundus abnormality, see the disease section page!

Signature of the doctor:

Page 1

BV1000S OCT binocular analysis report

AI-OCT was selected into the 8th batch of excellent Domestic medical equipment product list of China Medical Equipment Association

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AI-OCT was awarded the honorary title of "Scientific and Technological Progress in Ophthalmology of China Medical Equipment Association"

A cooperative research center for multimodal imaging AI in ophthalmology has been established with an authoritative eyehospital



Technical parameters

Chinese NMPA: Jiangxi 20202160485

Measuring characteristics	Axial resolution (in the tissue)： 5μm
	Horizontal resolution (in the tissue): 20μm
Scanning characteristics	The highest scanning speed : ≥80000 times / second
	Scan depth: 2.3mm
	Maximum scan range: 12mm * 9mm
Light source characteristics	Centre wavelength: 840nm
	Optical power: ≤750μW
	Refractive compensation range: -20D ~ + 25D

Scanning modes

Mode	Scanning ways	Physical size	Slice Direction
Large field	Single-line	12mm	Horizontal
	Six radial lines	9mm	Per 30°
	Large field	12mm x 9mm	horizontal or vertical
Glaucoma	Macular area	6mm x 6mm	Horizontal
	optic disc area	6mm x 6mm	Horizontal
Anterior segment	Single line	6mm	Horizontal
	Six radial lines	6mm	Per 30°

Note: When selecting the anterior segment mode, the peripheral devices of the anterior segment should be in place; the design and parameters are subject to change without prior notice.

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Bigvison Ophthalmic AI